B1 LoRA statute classifier: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Start with the in-house Phase 1 B1 representation and change a small low-rank part of the network for statute labels. This is the neural reference for later Phase 5 branches: it tests what the inherited representation can do before document-scale lexical evidence is added.

Architecture and mechanism

A frozen B1 backbone produces document-window representations. LoRA adds trainable low-rank updates to selected linear transformations; a multi-label head emits one logit per statute. Per-label decision thresholds are fitted on an official-train calibration split.

Measured result

Scored submission	1016271
Codabench macro-F1	0.2554
Archived local metric	0.2555 official-dev macro-F1

Its local official-dev macro-F1 is about 0.2555, close to the 0.2554 Codabench result. That agreement is evidence of a stable but limited baseline, not proof of convergence.

Limitations and next check

Long judgments can hide decisive statutory language outside the encoded windows; rare statutes make per-label thresholds noisy. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase5_b1_train.py, phase5_core.py, phase5_b1_cal_infer.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.